SOAR Playbook Development and Automation Report

Introduction

This practical focuses on designing and testing a **Security Orchestration, Automation, and Response (SOAR)** playbook to automate incident response for phishing-related threats.

The objective is to reduce manual effort by integrating detection, analysis, response, and incident management into a single automated workflow.

A phishing alert was simulated using **Wazuh**, validated with threat intelligence, mitigated using **CrowdSec**, and documented in **TheHive**.

Tools Used

Tool	Purpose
Splunk Phantom	Playbook automation
Wazuh	Alert generation
CrowdSec	IP blocking
TheHive	Case management

Objective of the Playbook

To automatically:

- Detect phishing alerts
- Extract suspicious IPs
- Validate IP reputation
- Block malicious IPs
- Create incident tickets

Playbook Workflow

The automated workflow is:

```
Wazuh Alert → Extract IP → Check Reputation → Decision → Block IP → Create Case
```

Simulate Phishing Alert (Wazuh)

Action:

- Open Wazuh dashboard
- Navigate to **Security Events / Alerts**

Simulation:

A phishing alert was generated with:

Field	Value
Alert Name	Suspicious File Download
Source IP	192.168.56.106
Severity	High

Outcome:

- Alert successfully triggered
- IP extracted for further analysis

Extract IP Address

Action:

- From Wazuh alert, identify source IP

Extracted IP:

```
192.168.1.100
```

Outcome:

- IP ready for threat intelligence check

Check IP Reputation

Action:

- Open AlienVault OTX
- Search for IP address

Observation:

- IP associated with suspicious/scanning activity
- Linked to known reconnaissance domains

Outcome:

- IP classified as **malicious/suspicious**

Decision Making

Logic:

- If IP is malicious → proceed to response
- If benign → stop workflow

Result:

- IP identified as malicious → continue

Block IP using CrowdSec

Action:

- Add IP to blocklist

Command (simulation/real):

```
cscli decisions add --ip 192.168.56.106
```

Outcome:

- IP successfully blocked
- Prevented further access

Create Incident Case in TheHive

Action:

- Open TheHive
- Create new case

Case Details:

Field	Value
Title	Phishing Alert – Suspicious IP
Severity	High
Observable	192.168.56.106

Outcome:

- Incident documented
- Case available for investigation

Automation using Splunk Phantom

Playbook Logic:

1. Trigger on Wazuh alert
2. Extract IP
3. Query threat intelligence
4. If malicious:
    - Block IP
    - Create case
5. Log action

Key Components:

- Filter block (alert detection)
- IP extraction function
- Threat intelligence integration
- Response actions (block + case creation)

Playbook Testing

The playbook was tested using a simulated phishing alert.

Test Results

Playbook Step	Status	Notes
Alert Trigger	Success	Wazuh alert detected

Playbook Step	Status	Notes
Check IP	Success	IP flagged as malicious
Block IP	Success	CrowdSec blocked 192.168.56.106
Create Case	Success	Case created in TheHive

Threat Analysis

The phishing alert indicated a suspicious file download activity from an external IP.

Threat intelligence confirmed that the IP had associations with scanning behavior, indicating potential attacker activity.

The automated playbook successfully:

- Detected the threat
- Blocked attacker access
- Documented the incident

Benefits of SOAR Implementation

- Faster response time
- Reduced manual workload
- Consistent incident handling
- Real-time threat mitigation
- Improved tracking and documentation

Playbook Summary

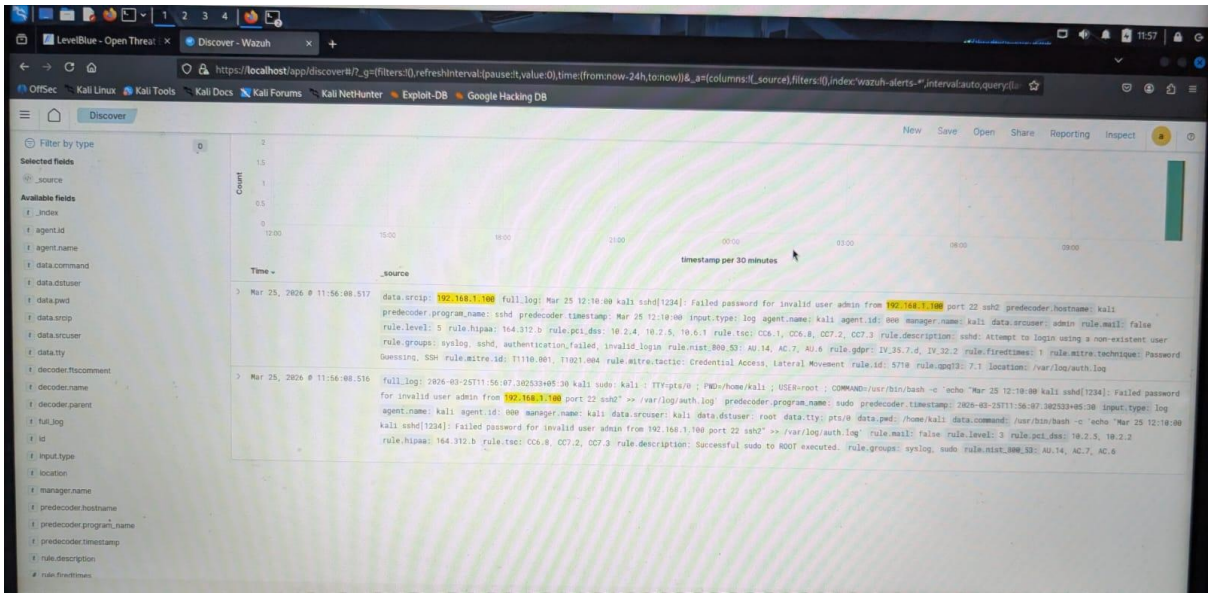
This playbook automates phishing incident response by extracting suspicious IPs from Wazuh alerts, validating them through threat intelligence, and blocking malicious sources using CrowdSec. It also creates incident cases in TheHive for tracking and investigation. The automation improves response speed, reduces manual effort, and ensures consistent security operations.

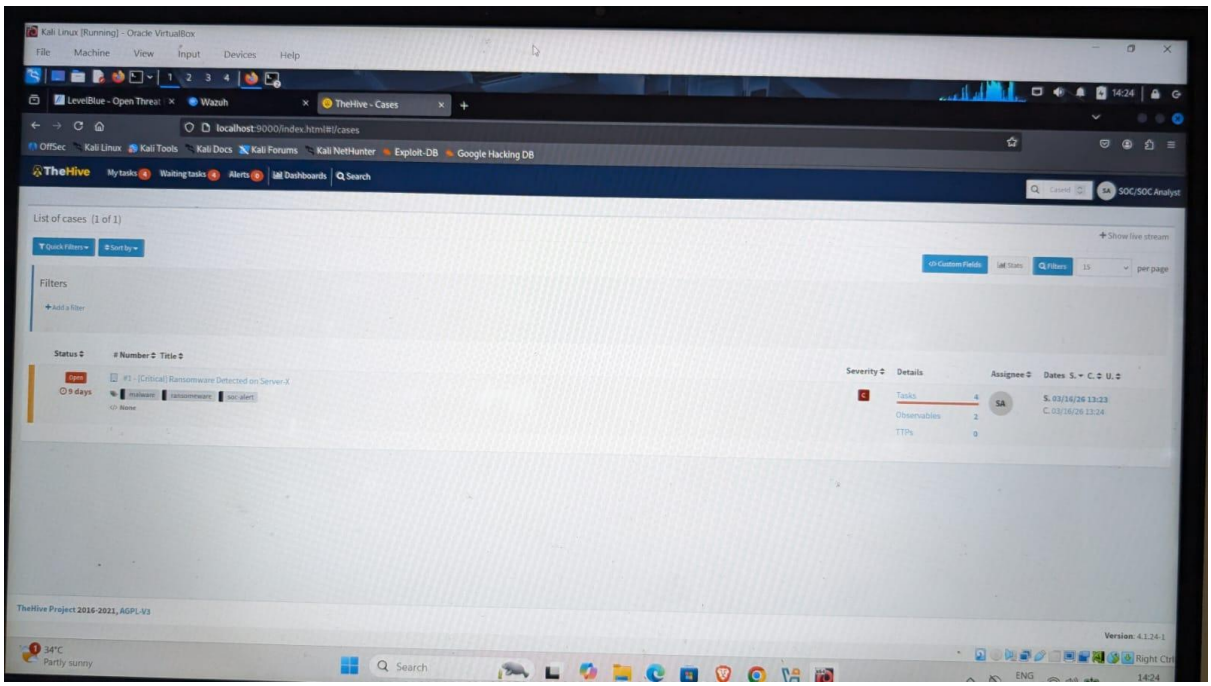
Conclusion

This practical demonstrates the design and implementation of a SOAR playbook for automated phishing incident response.

The integration of Wazuh, Splunk Phantom, CrowdSec, and TheHive enables efficient detection, validation, and mitigation of threats.

The successful execution highlights the importance of automation in modern SOC environments.





```
Session  Actions  Edit  View  Help                                   kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status crowdsec
○ crowdsec.service - Crowdsec agent
     Loaded: loaded (/usr/lib/systemd/system/crowdsec.service; disabled; preset: disabled)
     Active: inactive (dead)

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start crowdsec

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status crowdsec
● crowdsec.service - Crowdsec agent
     Loaded: loaded (/usr/lib/systemd/system/crowdsec.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-03-26 09:52:26 IST; 6s ago
 Invocation: c1ecf6b7657d4d63bfbf35fc67bfba96
    Process: 29004 ExecStartPre=/usr/bin/crowdsec -c /etc/crowdsec/config.yaml -t (code=exited, status=0/SUCCESS)
   Main PID: 29052 (crowdsec)
      Tasks: 9 (limit: 4546)
     Memory: 71.2M (peak: 72.7M)
        CPU: 3.581s
     CGroup: /system.slice/crowdsec.service
             ├─29052 /usr/bin/crowdsec -c /etc/crowdsec/config.yaml
             └─29081 journalctl --follow -n 0 _SYSTEMD_UNIT=ssh.service

Mar 26 09:52:22 kali systemd[1]: Starting crowdsec.service - Crowdsec agent...
Mar 26 09:52:26 kali systemd[1]: Started crowdsec.service - Crowdsec agent.

┌──(kali㉿kali)-[~]
└─$ sudo cscli decisions list
No active decisions

┌──(kali㉿kali)-[~]
└─$ sudo cscli decisions add --ip 192.168.56.106 --reason "Samba exploit attack"
INFO[26-03-2026 09:53:31] Decision successfully added

┌──(kali㉿kali)-[~]
└─$ sudo cscli decisions list
╭────┬────────┬──────────────────┬──────────────────┬────────┬─────────┬────┬────────┬────────────────┬──────
│ ID │ Source │ Scope:Value      │ Reason           │ Action │ Country │ AS │ Events │ expiration     │ Al
ert ID │
├────┼────────┼──────────────────┼──────────────────┼────────┼─────────┼────┼────────┼────────────────┼──────
│ 3  │ cscli  │ Ip:192.168.56.106 │ Samba exploit attack │ ban    │         │    │ 1      │ 3h59m56.101458032s │ 3
╰────┴────────┴──────────────────┴──────────────────┴────────┴─────────┴────┴────────┴────────────────┴──────

┌──(kali㉿kali)-[~]
└─$
```